Project 3: Stationarity diagnostic on a real series

Decomposition + ACF/PACF + Ljung-Box

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Context

This project builds on Project 2 (decomposition of `SouvenirSales.csv`) and asks you to assess the stationarity of the residual after log-decomposition. You will produce a complete stationarity diagnostic combining visual tools (ACF, PACF) and a formal test (Ljung–Box).

The point is to determine whether the residual is white noise (in which case decomposition is sufficient) or whether it carries exploitable autocorrelation that should be modelled with ARMA in Chapter 5.

Deliverables

- Reproducible `project3_<name>.py` with seed 42.
- Report PDF of 4–6 pages including all figures and a final recommendation.
- Optional 5-min oral defence.

Exercise 1 – Setup and decomposition

(1) Load `data/SouvenirSales.csv`, take logs.
(2) On $\log Y_t$, fit a model $\log Y_t = \beta_0 + \beta_1 t + \sum_{j=1}^{2} [a_j \cos(j\omega t) + b_j \sin(j\omega t)] + \varepsilon_t$ with $\omega = 2\pi/12$ by OLS.
(3) Compute the residual $\widehat{\varepsilon}_t$. Plot it versus t. Comment on its visual stationarity (mean stable around 0? variance constant?).

Exercise 2 – Sample autocorrelation

(1) Plot the sample ACF of $\widehat{\varepsilon}_t$ up to lag 24, with Bartlett bands.
(2) Plot the sample PACF up to lag 24.
(3) From the ACF/PACF patterns, propose a tentative ARMA order (p, q). Justify your choice using the signature table of Chapter 3.

Exercise 3 – Ljung–Box test

(1) Run the Ljung–Box test on $\widehat{\varepsilon}_t$ at lags $m \in \{6, 12, 18, 24\}$.
(2) Build a small table with columns: m, Q_{LB}, critical value $\chi^2_{m,0.95}$, p-value, decision.
(3) Conclude: is $\widehat{\varepsilon}_t$ white noise at the 5% level? Why does the answer typically differ between $m = 6$ and $m = 24$?

Exercise 4 – Comparison: pre-decomposition vs. post-decomposition

(1) Run the same Ljung–Box test on the raw $\log Y_t$ (no decomposition) at the same lags. What happens?
(2) Compute the variance ratio $\mathrm{Var}[\widehat{\varepsilon}]/\mathrm{Var}[\log Y]$. What does it tell you about the explanatory power of the decomposition?
(3) Discuss in 5 lines: even if the residual fails the Ljung–Box test, the decomposition is still useful — why?

Exercise 5 – Bonus: AR(1) modelling of the residual

If the Ljung–Box test rejects white noise, fit an AR(1) on $\widehat{\varepsilon}_t$:

$$\widehat{\varepsilon}_t = \phi\,\widehat{\varepsilon}_{t-1} + \eta_t.$$

(1) Estimate $\widehat{\phi}$ by OLS.
(2) Plot the residuals $\widehat{\eta}_t$ and re-run the Ljung–Box test.
(3) Has the AR(1) absorbed the remaining autocorrelation?

Marking grid (out of 100)

Criterion	**Points**
OLS decomposition + residual plot	15
ACF / PACF visualisation + Box-Jenkins reading	25
Ljung-Box table + interpretation	25
Comparison pre/post + variance reduction	15
Bonus AR(1) fit + residual diagnostics	10
Reproducibility + report quality	10
Total	**100**

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